

Nathaniel W. Brooks

Software Engineer

Education

- ❖ **TOWSON UNIVERSITY**
COMPUTER SCIENCE - 3.8 GPA
AUG 2022 - MAY 2026

RELEVANT COURSEWORK:

- DATA STRUCTURES/ALGORITHM ANALYSIS
- SOFTWARE ENGINEERING
- DATA COMMUNICATIONS & NETWORKING
- OBJECT-ORIENTED DESIGN & PROGRAMMING
- OPERATING SYSTEMS
- DATABASE MANAGEMENT SYSTEMS
- COMPUTER GRAPHICS
- ARTIFICIAL INTELLIGENCE
- MOBILE APPLICATION DEVELOPMENT
- WEB-BASED PROGRAMMING
- PROGRAMMING LANGUAGES DESIGN & IMPLEMENTATION

- ❖ **CARROLL COUNTY CAREER AND TECHNOLOGY CENTER**
CISCO CYBER OPERATIONS - 4.0 GPA
SEP 2021 - MAY 2022

Formalized introduction to the characteristics of cyber-crime, security principles, technologies, and procedures to defend networks. Learned how to use Python to create programs that would interact with the CISCO architecture, and how computer programs are executed.

Work Experience

- ❖ **FORTINET FEDERAL - PRODUCT MANAGEMENT & CLOUD INTERN**
JUN 2025 - AUG 2025

Architected and developed a policy compliant full-stack automation platform for FortiGate configuration and isolated network provisioning using Terraform, FastAPI, and Svelte. Led system design, UI/UX implementation, and infrastructure-as-code workflows while self-learning new technologies; gained hands-on experience with vSphere, Fortinet products, and federal compliance frameworks (FedRAMP, ZTNA, SASE, CMMC, NIST).

- ❖ **KITE TECHNOLOGY GROUP - SECURITY & OPERATIONS INTERN**
MAY 2024 - MAY 2025

Managed the NOC board by addressing cybersecurity alerts to ensure network security. Supported audits by deploying training campaigns, using M365 tools (Azure, Intune, Entra), and coordinating with clients on device management.

Projects

- ❖ **LINGANORE UNITED METHODIST CHURCH WEBSITE**
SPRING 2025

Solo developer for a full-stack church website with a custom admin panel and end-to-end control over the technology stack. Opted against third-party CMS and authentication services in favor of building bespoke systems tailored to exact needs. Built with Vite+React, ExpressJS, and MySQL, utilizing RESTful APIs.

- ❖ **JOUST - GAME**
SPRING 2025

JOUST was developed for my Computer Graphics course as a physics-based, pixel-stylized 3D jousting game using Three.js, as my first formal introduction to GLSL shaders and WebGL. The game features dynamic camera transitions, a points-based scoring system, and real-time collision mechanics with stylized knight animations.

- ❖ **CAPTION CREATORS - WEB APPLICATION/GAME**
2024

Semester long project in my Software Engineering course. An online round-based multiplayer game using AI to generate prompts and interact with YouTube API to propagate videos for players to caption, with gameplay similar to that of Quiplash from Jackbox Games. Built with NextJS and MySQL.

Certifications

❖ **JAVA SE 8 ORACLE CERTIFIED ASSOCIATE**

2023 - 1Z0-808

Gained proficiency in Java by building real-world applications, including object-oriented designs, algorithms, and data structures. This certification demonstrates a deep understanding of Java fundamentals, including exception handling, multithreading, and working with APIs.

Coding Languages and Tools

❖ **JAVA**

I fell in love with object-oriented programming during my computer science classes, where Java became my strongest skill due to its consistent focus in the curriculum. This led me to create several applications. My passion for OOP also pushed me to earn an Oracle Certified Associate Java Programmer certification, to further solidify my expertise.

❖ **PYTHON**

I first learned Python at the Career and Technology Center, where I built various tools and applications that deepened my understanding of the language. I have used Python on and off for several years, and recently it has become my go-to for backend APIs.

❖ **FIGMA**

I love UI/UX design and often start by sketching and wireframing interfaces before building the actual product. Bringing ideas to life through thoughtful design is one of my favorite parts of the process.

❖ **REACT | SVELTE | HTML/CSS | JAVASCRIPT**

I build modern, reusable UI components for highly interactive frontends, including real-time web multiplayer applications that call for extra technical creativity. Redesigned several websites to improve responsiveness, performance, and overall user experience, boosting load times and visibility. I focus on efficiency, accessibility, and SEO to ensure products feel fast and polished on any device.

❖ **TERRAFORM**

Picked up Terraform on the fly and built a dynamic frontend-driven system that generates live Terraform JSON configurations enabling automated infrastructure updates, resource provisioning, and seamless integration with an infrastructure-as-code workflow.

Skills

❖ **ADAPTABILITY**

I've consistently shown the ability to quickly grasp new technologies by building on a strong understanding of fundamentals. In school, I learned the MARIE machine architecture in just one day to prepare for an exam, which I aced. At Kite Technology Group, I quickly adapted to M365 tools like Intune and Entra, along with several other platforms to help meet company needs. I am very confident in my ability to effectively learn and contribute to any team or project I'm part of.

❖ **COMMUNICATION**

During my internship at Kite Technology Group, I worked directly with clients during on-site engagements and earned a reputation for being reliable and easy to work with. At Fortinet Federal, I integrated quickly with the product management team and contributed meaningfully while learning from more senior colleagues. Outside of work, I love meeting new people and building strong relationships.

❖ **PROBLEM SOLVING**

When faced with programming problems, I'm relentless in finding solutions, often thinking about them non-stop, sometimes skipping meals or staying up until the early morning to see them through to completion. I actively seek out projects where there's a knowledge gap; allowing me to learn, grow, and expand my development toolkit.